

Music, Motion, and Mixed Reality: An Interdisciplinary, Problem-Based Educational Experience

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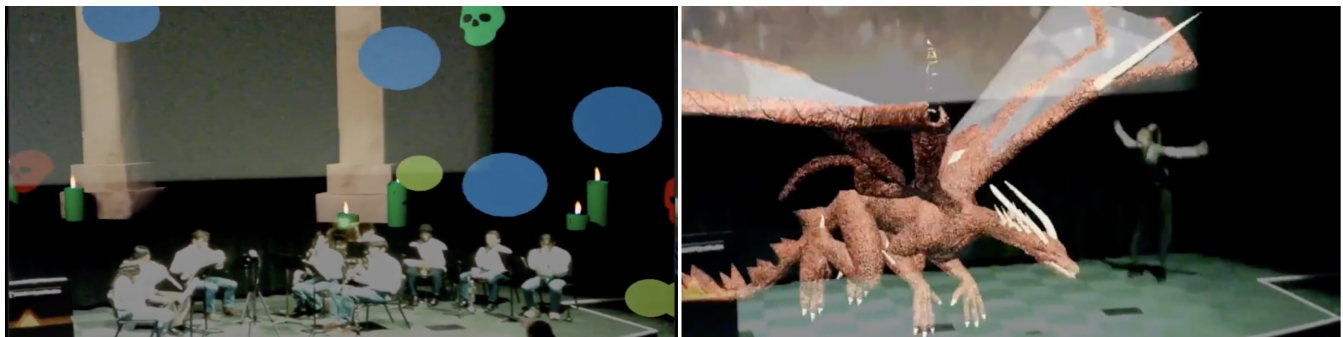


Figure 1: Live AR enhanced performances created by students as seen by audience members wearing Magic Leap One AR Devices – (a) Catacomb scene for *Pictures at an AR Exhibition* – (b) Slaying a virtual dragon for *Singring and the Glass Guitar*

ABSTRACT

This talk describes an interdisciplinary educational experience involving cohorts of students studying Computer Science, Dance, 3D Digital Design and Music. The experience takes a problem based learning approach towards a target goal of presentation of a set of live performances that combine music and dance with augmented reality. Our overall approach of integration of the varying disciplines as well as impressions of students involved in the experience is discussed.

CCS CONCEPTS

• **Applied computing** → **Media arts; Performing arts**; • **Computing methodologies** → **Mixed / augmented reality**; • **Social and professional topics** → **Computing education**.

KEYWORDS

live performance, augmented reality, problem based learning, interdisciplinary curriculum

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1 INTRODUCTION

A recent trend in higher education is the use of large interdisciplinary projects to provide collaborative and multidisciplinary experiences for students [Amoo et al. 2020]. In this talk, we describe one such experience: a set of interdisciplinary educational performances involving faculty and students in computer science, 3D design, music and dance designed as a joint learning exercise. We employ a problem based learning (PBL) approach. PBL is an experiential methodology where emphasis is placed on students developing a solution of a real world problem [Gallagher et al. 1992] which has been shown to notably improve student soft skills such as teamwork, responsibility, and communication. This is particularly true in the context of Computer Graphics education [Martí et al. 2006].

Our overarching project addresses the use of augmented reality technology to enhance live performance. The interdisciplinary nature of our problem encourages teamwork and communication not only within teams but also between teams and disciplines. The focus on live performance only sharpens these skills [McDonald and

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Table 1: Students Courses and Responsibilities

Discipline	Course/Learning Activity	Student Responsibilities
3D Design	Independent Studies	Asset design and creation
Computer Science	Course: Applications in VR	VR App development, network architecture & communication, music analysis
Dance	Course: Dance and the Metaverse	Choreography and dance performance
Music	Wind and Brass Ensembles	Live musical performance

Wolfe 2008; Schweppe and Geigel 2010]. The talk will highlight the project, our educational approach, and present student reactions to the experience.

2 THE PROJECTS

For the student experience, we set out to realize the following two projects, each focusing on the use of Augmented Reality to enhance live performance.

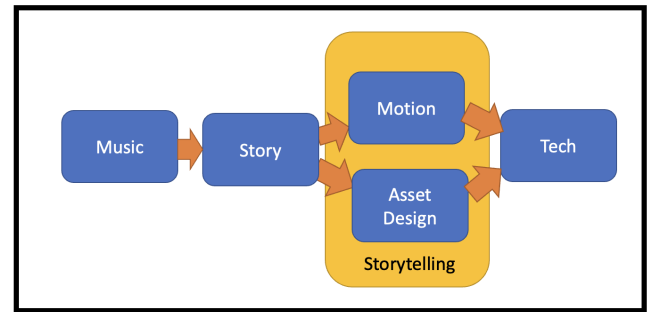
- *Pictures at an AR Exhibition* – Live orchestral performance of several movements of Mussorgsky's *Pictures at an Exhibition*. Using AR, the musicians are placed in immersive environments with effects based on analysis of the live music.
- *Singring and the Glass Guitar* – A live dance presentation of the 18-minute rock opus by Utopia. The song is an “electrified faiytale” featuring a group of explorers who must battle the four elements to find four keys and save the spirit of Harmony.

Both pieces were performed live at the 2022 Frameless XR Showcase at the Rochester Institute of Technology in November of 2022¹ (Figure 1). In both cases, Magic Leap One AR devices are worn by audience members to immerse viewers in a given digital and dynamic environment during performance.

3 APPROACH

Students studying Computer Science, 3D Digital Design, Dance, and Music, were involved in the experience. Each student was enrolled in a course or learning activity corresponding their academic focus (Table 1). Courses were scheduled at a common time to encourage interaction between teams. Students were expected to contribute their particular expertise to the project and instructors served as facilitators and directors (artistic, dance, technical) guiding the overall direction of the project.

We approached the the project using the underlying workflow illustrated in Figure 2 to emphasize the interconnected nature of all the disciplines in creation of the final performance. Each performance is driven by story and for both, the music dictated the story, either explicitly (e.g. *Singring* where the lyrics convey the story) or less directly (e.g. For *Pictures* where stories were based on the artworks that inspired the music). Once the story was set, students in the dance and design classes defined and created choreography and dynamic 3D assets respectively, that were used to tell each story. Finally, based on the storytelling mechanisms defined, the tech team created the applications that are deployed on the AR devices that audience members use to correctly view the digital assets in the physical performance space.

**Figure 2: Creative workflow of live productions**

4 STUDENT REACTIONS

At the end of the term, students were asked to provide feedback on their experiences with working on the projects. As expected of a PBL method, students took a more active role in their own learning; within each team, they defined their own tasks and learned what they needed to know in order to complete the task at hand which often required independent exploration.

Students expressed much curiosity about expertise of students in other domains. Though not expected to become experts in these other domains, students refined approaches in their own domains based on the experiences of those in others. Given the interconnected nature of the team tasks, communication between students between teams became essential and guided students with the work required by their own respective teams.

The “live” aspects of the project (the fact that the project would be shown to a live audience), notably increased student motivation. This is especially true as the time to presentation came near as students did whatever it took to assure that the *show would go on* when scheduled to do so.

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¹<https://www.rit.edu/framelesslabs/symposium-2022>